

“E-commerce customer dataset”

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Introduction

Project overview:

This project focuses on cleaning and improving the quality of a customer dataset containing 61 records. The dataset includes customer demographic information, purchasing behavior, membership details, payment methods, and satisfaction scores.

The objective of this project was to identify and resolve data quality issues that could affect the accuracy and reliability of future analysis. During the cleaning process, several types of inconsistencies were detected, including incorrect gender entries, invalid and missing age values, inaccurate total spending calculations, business rule violations in returned items, and inappropriate data types.

The dataset was cleaned using Python and the Pandas library by validating records against logical business rules, correcting data entry errors where reliable corrections could be made, handling missing values appropriately, and optimizing data types for improved efficiency.

The final result is a clean, consistent, and analysis-ready dataset that can be used for exploratory data analysis, reporting, or machine learning applications.

Dataset Description

Column	Data Type	Description
customer_id	Integer	Unique identifier assigned to each customer.
first_name	String	Customer's first name.
gender	Category	Customer's gender (M = Male, F = Female).
age	Integer	Customer's age in years.
city	Category	City where the customer resides.
province	Category	Province where the customer resides.
signup_date	Date	Date when the customer registered.
membership_tier	Category	Customer's membership level (e.g., Silver, Gold, VIP).
purchase_count	Integer	Total number of purchases made by the customer.
avg_order_value	Float	Average monetary value of each purchase.
total_spending	Float	Total amount spent by the customer across all purchases.
last_purchase_days	Integer	Number of days since the customer's most recent purchase.
payment_method	Category	Customer's preferred payment method.
device	Category	Device used to make purchases (e.g., Android, iPhone, Web).
discount_used	Category	Indicates whether the customer used a discount (Yes or No).
returned_items	Integer	Number of items returned by the customer.

satisfaction_score	Integer	Customer satisfaction rating on a scale from 1 (lowest) to 5 (highest).
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Data Exploration

Initial exploration was performed using Pandas functions including:

- Head()
- Info()
- Describe()
- Unique()

These analysis provided and overview of the dataset structure and helped identify potential quality issues

Identified Data Quality issues

The following problems were detected:

- Duplicated rows
There was one duplicated row with identical information
 - Missing values
Age & total spending
 - Business rule violation
Several records had incorrect Total Spending values.
Six records had more returned items than purchased items, including one customer with 0 purchases but 2 returned items.
 - Consistency issue
Gender inconsistent with first name
 - Data type optimization and correction
Column for date had the wrong data type. Object columns that could be optimized.
 - Outlier
One customer had an age of 145.
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Data cleaning process

The dataset was analyzed to identify data quality issues, including inconsistent values, missing data, invalid entries, duplicate records, and business rule violations. The following steps were performed during the cleaning process:

Inconsistent Gender Values

The relationship between first_name and gender was checked to identify mismatched entries. The incorrect gender values were corrected only when the expected value could be determined with confidence.

Missing and Invalid Age Values

The age column was inspected for missing and unrealistic values. One missing age value was replaced using the median age of the dataset. Additionally, an invalid age value of 145 was corrected to 45 after identifying it as a likely data entry error.

Incorrect Total Spending Values

The total_spending column was validated using the relationship between purchase_count and avg_order_value. Records with incorrect total spending values were recalculated using the formula:

Total Spending = Purchase Count × Average Order Value

Invalid Returned Items Values

The returned_items column was checked against the business rule that returned items cannot exceed purchased items. Six records violated this rule, including one record with returned items despite having no purchases. Since the correct values could not be reliably determined, these invalid values were replaced with missing values.

Duplicate Records

The dataset was checked for duplicate rows and duplicate customer profiles. One exact duplicate record was identified and removed to prevent repeated information from affecting future analysis.

Data Type Optimization

Column data types were optimized to improve consistency and efficiency. Categorical variables were converted to category type, text columns were converted to string format, and date columns were converted to datetime format.

After completing these steps, the dataset was cleaned and prepared for further analysis.

Tools used

- Jupyter notebook
- Python
- Pandas
- Numpy

Conclusion

This project focused on improving the quality and reliability of a customer dataset by identifying and resolving various data quality issues. Through the cleaning process, inconsistencies, invalid values, missing data, duplicate records, and business rule violations were investigated and handled using appropriate methods.

The final dataset is now more accurate, consistent, and structured, making it suitable for further analysis and visualization. This project also highlighted the importance of understanding relationships between different columns and making careful decisions when correcting real-world data issues.

By documenting each cleaning step and the reasoning behind each decision, the dataset became more reliable and ready to support future customer behavior analysis and business insights.